PROJECT JÖRMUNGANDR

Consolidated Master Document

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Document Type: Complete Systems Architecture & Implementation Framework
Status: Complete – Ready for Implementation

AI tools were used as research and editing assistance. All concepts, design decisions, and strategic choices are the author's responsibility. The named AI collaborator on this document is ***Corvid****. This work draws on publicly available data, peer-reviewed literature, and industry reports. Sources are cited throughout and listed in the Bibliography.*

Contents

1 M0: EXECUTIVE SUMMARY & VISION

Project Jörmungandr is a modular, closed-loop oceanic regeneration architecture designed to restore major oceanic basins through autonomous surface nodes, deep-sea static structures, and mobile mid-water drones.

Name Justification

Jörmungandr (The Midgard Serpent) encircles the planet, consuming its own tail—representing the closed-loop circular economy. The architecture rides ocean currents, consumes pollution, and returns refined materials to the cycle. The name is structurally and mythologically aligned with the design philosophy.

Core Philosophy

Project Jörmungandr is a **public utility** for ocean restoration. It is designed as permanent infrastructure for planetary repair, not a business venture. Like a municipal water treatment plant or a national highway system, it is funded by public and philanthropic sources to provide a continuous public good. Cost offset comes from material sales (recycled plastics, metals), carbon credits, and a **Sustainability Stamp** licensing program—all of which reduce the burden on public funding but are not expected to achieve profitability.

Phased Objectives

- **Phase 0:** Validates core technology and proves the concept works.
- **Phase 1:** Demonstrates measurable ecological impact at basin-scale (North Pacific Gyre) while scaling systems for operational viability.
- **Phase 2:** Scales systems to global deployment, building the infrastructure for planetary-scale regeneration.
- **Phase 3+:** The long-term vision for complete ocean regeneration—global fleet operations, fusion-powered platforms, and true basin-scale impact.

2 M1: SYSTEM ARCHITECTURE

2.1 1.1 Component Table

Tier	Component	Primary Function	Energy	Governance
Surface (Passive)	Bio-Bubble (Cellular)	Algal filtration + macroplastic capture	Solar (passive)	Oceanic Trust
Surface (Active)	Skin Skimmer	Macroplastic trawling (neuston layer)	Solar-Electric	Oceanic Trust
Mid-Water	Gandr Drone	Active mid-water plume suction	Battery (swapped)	Oceanic Trust
Mid-Water (Hub)	Micro-Hive	Battery swap, tank dump, data relay	Hybrid (Solar/Diesel)	Oceanic Trust
Benthic	SSS Crawler	Sediment vacuum (top 10cm)	Umbilical (from Barge)	Oceanic Trust
Benthic (Support)	SSS Barge	Power, filter swap, slurry storage	Diesel / Solar	Oceanic Trust
Governance	PAP Protocol	Polluter identification & transparency	Data/Reputation	Oceanic Trust

2.2 1.2 Material Flow Summary

Collection Stage:

- **Skin Skimmers** collect macroplastics from the surface neuston layer
- **Gandr Drones** collect microplastics from the mid-water column (10-200m)
- **SSS Crawlers** collect heavy metal-contaminated sediment from the seabed (top 10cm)
- **Bio-Bubbles** absorb dissolved nutrients, CO_2, and microplastics via algal uptake

Processing Stage:

- **Macroplastics** are sorted by polymer type (PE, PP, PET) and mechanically recycled
- **Microplastics** are concentrated via pyrolysis into crude oil or fuel
- **Heavy Metals** are extracted via electro-winning into pure metal products
- **Algal Biomass** is processed via anaerobic digestion into methane and biofertilizer

Output Stage:

- **Recycled PET** returns to the economy as new products

- **Crude Oil** powers operations or enters fuel markets
- **Pure Metals** (Pb, Cd, rare earths) are sold to industry
- **Methane** generates energy for the Micro-Hive and processing facilities
- **Carbon Credits** are generated from sequestered CO_2

2.3 1.3 Data Flow Summary

Internal Data Sources:

- Skin Skimmers: GPS location, collection volume, polymer type (FTIR)
- Gandr Drones: Depth profile, microplastic concentration, water quality (sonde)
- SSS Crawlers: Sediment metal concentration, location, volume extracted
- Bio-Bubbles: Growth rate, uptake efficiency, water quality (pre/post)
- Active Telemetry: pH, temperature, DO, conductivity, turbidity (continuous)

External Data Sources:

- AOMI (Atlas of Ocean Microplastics): Global microplastic concentration data
- EMODnet: Marine litter, chemistry, and habitat data
- Argo Floats: Temperature, salinity, and dissolved oxygen profiles
- Copernicus Marine Service: Current forecasts and drift modeling

Data Processing:

- Harmonization layer applies common standards (FAIR principles)
- Cross-validation compares passive (lab) vs. active (sensor) data
- Correction factors applied to active data based on passive validation

Data Outputs:

- PAP Database: Unified dataset with source attribution
- Public Dashboard: Heat maps, timelines, company attribution
- Data Request Portal: Custom reports for researchers and journalists

3 M2: BIO-BUBBLE SYSTEM

3.1 2.1 Layer Structure (Inside → Out)

Layer	Name	Material	Opening Size	Function
1 (Core)	Mitochondria Bag	Fine mesh (monofilament PET)	50µm	Holds single algal strain; kill switch inside
2	Containment Bubble	Fine mesh (monofilament PET)	50µm	Primary algae retention
3	Macro-Filter	Coarse woven mesh	5mm	Captures bottle caps, rings, large fragments
4	Air Bladder Spacers	Welded PVC/TPU	N/A	Maintains 15-30cm separation between layers
5 (Outer)	Security Mesh	High-tensile Dyneema/PET	10cm grid	Impact protection; mooring attachment point

3.2 2.2 Strain Selection (Phase 0)

Strain	Type	Uptake Target	Survivability	Feasibility
Nannochloropsis	Green Algae	Microplastics, Nitrogen	High (temperate)	9/10
Phaeodactylum	Diatom (Pennate)	Heavy Metals (Pb, Cd)	High (cosmopolitan)	9/10
Synechococcus	Cyanobacteria	Trace Metals	Very High (gyre-native)	10/10
Tetraselmis	Green Algae	Nutrients, Plastics	High (estuarine)	8/10

3.3 2.3 Harvest Strategy

Cohort	Strains	Number of Bubbles	Harvest Frequency	Purpose
Fast Harvest (4 cohorts)	All four	16 micro-bubbles (4 per strain)	Monthly (staggered weekly)	Rapid data generation
Long-Term Study	All four	4 sentinel bubbles (1 per strain)	Quarterly	Max growth, thermal effects

3.4 2.4 Kill Switch (Per Mitochondria Bag)

Component	Trigger	Action	Redundancy
UV-C LED Array	Breach detected or remote command	Emits 254nm UV-C (lethal to algae)	Dual arrays (redundant power)
Peroxide Capsule	UV-C failure	Releases mild oxidant (strain-specific)	Backup chemical
Bag Pressure Sensor	Continuous monitoring	Detects tear or overgrowth; triggers kill	3 independent sensors

3.5 2.5 Phase 0 → Phase 2 Scaling

Phase	Bubble Size	Diameter	Number	Purpose
Phase 0	30-50m^2	~6-8m	4-8	Validate design, strain, kill switch, filtration efficiency
Phase 1	100m^2	~11m	10-20	Validate drift management, data collection, CO_2 drawdown metrics
Phase 2	**1,000m^2**	~35m	50-100	Meaningful regional impact—process significant water volumes

Rationale for 1,000m² Phase 2 Size:

- Proven engineering (soft membrane structure, not rigid)
- Meaningful impact: 18,688 tons CO_2/year per bubble (at high-performance microalgae rates)
- Cost-effective scaling: $50-100M per unit, 8x more cost-effective than 100m bubbles
- Feasible to deploy with existing vessels

3.6 2.6 Drift Management Strategy

Bio-Bubbles drift with ocean currents. This is intended—it allows passive movement through the water column. However, drift must be managed to prevent dispersal into shipping lanes or sensitive ecosystems.

Method	Implementation	Purpose
GPS Tracking	Each bubble is equipped with a GPS transponder (Iridium satellite)	Real-time position monitoring
Drogue	A 2m diameter fabric cone is attached to the bubble	Prevents wind-driven drift (keeps bubble in water column)
Scheduled Repositioning	Support vessel intercepts and relocates bubbles that drift outside acceptable boundaries	Maintains bubble distribution
Active Positioning (Future)	Small solar-powered thrusters on each bubble	Long-term goal for autonomous positioning

Acceptable Drift Boundaries:

Zone	Boundary	Action
Primary Zone	Within designated deployment area	No action
Secondary Zone	Approaching shipping lanes or sensitive ecosystems	Schedule repositioning
Emergency Zone	Within 10km of shipping lane or protected area	Immediate repositioning

4 M3: GANDR DRONE & MICRO-HIVE

4.1 3.1 Gandr Drone Specifications

Parameter	Phase 0	Phase 1	Rationale
Length	3.0m	3.5m	Larger tank capacity
Diameter	0.8m	0.9m	Improved hydrodynamics
Dry Mass	500kg	550kg	Additional sensors
Wet Mass (full tank)	1,100kg	1,300kg	Larger tank capacity
Operating Depth	10-200m	10-200m	Proven range
Mission Duration	6 hours	**12-16 hours**	Battery upgrade
Sludge Tank	600kg	800kg	Larger capacity
Battery Pack	50kWh Li-ion	**100kWh Li-ion**	Doubled capacity

4.2 3.2 Micro-Hive Specifications

Parameter	Phase 0	Phase 1
Type	Semi-submersible barge	Semi-submersible barge
Length	20m	25m
Beam	10m	12m
Docking Ports	5	8
Battery Storage	10 packs	20 packs
Sludge Storage	30,000L	100,000L
Power (solar)	20kW	40kW
Power (diesel backup)	100kW	150kW

4.3 3.3 Docking Procedure (Human-Performed)

Step	Action	Duration	Operator
1	Drone approaches hive (acoustic beacon)	—	Autonomous
2	Docking arm guides drone into port	2 minutes	Autonomous
3	Capture mechanism secures drone	1 minute	Autonomous
4	Technician disconnects full sludge tank	3 minutes	Human
5	Technician connects empty tank	3 minutes	Human
6	Technician swaps battery pack	3 minutes	Human
7	Drone releases and departs	1 minute	Autonomous
8	Full tank pumped to hive storage	2 minutes	Automated

4.4 3.4 Fleet Management (Phase 0)

Drone ID	Primary Role	Status
Gandr-1	Active (primary)	Online
Gandr-2	Active (secondary)	Online
Gandr-3	Active (tertiary)	Online
Gandr-4	Docking/Maintenance	Offline (rotating)
Gandr-5	Cold Spare	Offline (standby)

4.5 3.5 Phase 0 → Phase 2 Scaling

Aspect	Phase 0	Phase 1	Phase 2
Drone Quantity	5	12	100
Mission Duration	6 hours	12-16 hours	24+ hours
Tank Capacity	600kg	800kg	1,500kg
Autonomy Level	Level 1-2	Level 3-4	Level 4-5
Docking Ports	5	8	20
Sludge Storage	30,000L	100,000L	500,000L

5 M4: SSS CRAWLER & BARGE

5.1 4.1 Canister Vacuum Model

Component	Canister Analog	Function
SSS Barge	Wall Outlet	Power, communication, sludge storage
Umbilical	Power Cord	Power + Data + Compacted sludge return
Benthic Garage	Canister	Filtration, compaction, water reintegration
Vacuum Head ROV	Motorized Head	Inlet, brushes, agitation, mobility

5.2 4.2 Garage (Benthic Processing Unit)

Parameter	Phase 0	Phase 2
Length	2.0m	4.0m
Width	1.5m	3.0m
Height	1.5m	2.0m
Mass (dry)	500kg	3,000kg
Depth Rating	100m	4,000m
Pressure Housing	Aluminum	Titanium/composite

5.3 4.3 Vacuum Head ROV

Parameter	Phase 0	Phase 2
Length	1.5m	2.0m
Width	1.0m	1.5m
Mass (dry)	150kg	300kg
Brush Head	1.0m width	1.5m width
Vacuum Intake	100mm diameter	150mm diameter
Depth Rating	100m	4,000m

5.4 4.4 Umbilical Specifications

Parameter	Phase 0 (Barge→Garage)	Phase 2 (Barge→Garage)
Length	150m	5,000m
Power Conductors	3x $10mm^2$	6x $10mm^2$ (redundant)
Data Fibers	2x SMF	4x SMF (redundant)
Sludge Return	1x 25mm hose	2x 25mm hose (redundant)
Outer Sheath	Polyurethane	Polyurethane + steel armor

5.5 4.5 Phase 0 → Phase 2 Scaling

Aspect	Phase 0	Phase 2
Depth Rating	100m	4,000m
Crawler Size	2m length, 500kg	3m length, 1,500kg
Umbilical Length	150m	5,000m
Umbilical Type	Polyurethane	Steel-armored
Barge Type	Anchored barge	DP vessel
Slurry Storage	20,000L	100,000L
Number of Crawlers	1	10
Status	Operational prototype	R&D + limited operational deployment

Rationale for Phase 2 Deployment:

- Deep-sea sediment is not actively polluting like surface and mid-water pollutants
- "Put out the fire first, then treat the burn"
- SSS is a long-term R&D effort, not a Phase 1 operational priority

6 M5: SKIN SKIMMER

6.1 5.1 Specifications

Parameter	Phase 0	Phase 1	Phase 2
Type	Twin-hulled catamaran	Twin-hulled catamaran	Twin-hulled catamaran
Length	**5m (bay vessel)**	**15-20m (ocean-going)**	20m+
Beam	2.5m	4m	6m
Speed (cruise)	2 knots	2 knots	2 knots
Speed (transit)	4 knots	5 knots	6 knots
Propulsion	Electric (solar-charged)	Diesel-electric hybrid	Diesel-electric hybrid
Net Width	5m	8m	10m+
Collection Bin	1,000L	5,000L	10,000L+
Quantity	2	5	50
Autonomy	Semi-autonomous	Semi-autonomous	Fully autonomous

Rationale:

- Phase 0 uses bay vessels for process testing (cost-effective, safe)
- Phase 1 uses full-size ocean-going vessels (15-20m) with proper sea-keeping capability
- Phase 2 scales to 50 vessels for global impact

6.2 5.2 Operational Logistics

Phase	Processing Location	Method
Phase 0	Shore (Commercial MRF)	Standard commingled recycling
Phase 1	Offshore Sorting Vessel + Land Hub	Sorted/Cubed (commodity grade)
Phase 2+	Offshore + Distributed Hubs	Full vertical integration

6.3 5.3 Phase 0 Cost

Component	Specification	Quantity	Unit Cost	Total
Vessel	5m catamaran	2	$35,000	$70,000
Trawl Net	5m width	2	$8,000	$16,000
Collection Bins	1,000L RFID-tracked	6	$1,500	$9,000
Sensor Suite	GPS, camera, AIS	2	$5,000	$10,000
Dock Logistics	Crane, trucking	1	$15,000	$15,000
Tipping Fees	MRF processing	100 tons	$50/ton	$5,000
Subtotal				**$125,000**
Contingency (30%)				**$37,500**
TOTAL				**$162,500**

7 M6: MONITORING & TELEMETRY

7.1 6.1 Dual-Track Validation

Track	Method	Purpose
Track A (Passive)	Manual sampling, lab analysis	Establishes baseline; validates active sensors
Track B (Active)	Continuous telemetry (sondes, counters)	Captures causality (temperature, current, salinity effects)
Cross-Validation	Compare Track A vs. Track B	Identify systematic biases; apply correction factors

7.2 6.2 Phase 0 Station Layout Summary

Upstream (Pre-Stations):

- Pre-Station 1 (Passive): Niskin bottles at 1m and 5m; manual logging; weekly sampling
- Pre-Station 2 (Active): Multi-parameter sonde (pH, temp, DO, conductivity, turbidity); ADCP (current speed/direction); telemetry relay to Micro-Hive; continuous monitoring (10-minute intervals)

Bio-Bubble Web: Located between Pre and Post stations; 16 micro-bubbles + 4 sentinel bubbles; anchored in coastal waters
Downstream (Post-Stations):

- Post-Station 1 (Passive): Niskin bottles at 1m and 5m; manual logging; weekly sampling
- Post-Station 2 (Active): Multi-parameter sonde; in-line microplastic counter (FTIR-based); telemetry relay to Micro-Hive; continuous monitoring

Data Relay:

- Micro-Hive collects data from all active stations via acoustic modem (short range) and satellite uplink (global)
- Shore-based data aggregation and cross-validation with passive samples

7.3 6.3 Active Telemetry Equipment

Item	Specification	Cost (approx.)
Multi-parameter Sonde	YSI EXO2	$7,300
ADCP	Acoustic Doppler Current Profiler	$25,000
In-line Microplastic Counter	FTIR-based	$40,000
Solar Power System	100W + battery	$1,500
Iridium Satellite Modem	RockBLOCK	$1,000
Data Logger	Custom/Commercial	$2,000

7.4 6.4 External Data Integration

Dataset	Source	Use
Microplastic Concentration	AOMI	Baseline comparison; site selection
Marine Litter	EMODnet	Validate collection data
Temperature/Salinity/DO	Argo floats	Environmental context
Ocean Currents	Copernicus	Drift modeling; route planning
Heavy Metals	EMODnet Chemistry	Baseline for SSS targets

7.5 6.5 Phase 0 Cost

Component	Specification	Quantity	Unit Cost	Total
Niskin Bottles	5L, oceanographic	8	$800	$6,400
Handheld Sonde	YSI ProDSS	1	$5,000	$5,000
Multi-parameter Sonde	YSI EXO2	4	$7,300	$29,200
ADCP	Acoustic Doppler	2	$25,000	$50,000
In-line Counter	FTIR-based	1	$40,000	$40,000
Solar Power System	100W + battery	4	$1,500	$6,000
Iridium Modem	RockBLOCK	4	$1,000	$4,000
Data Logger	Custom/Commercial	4	$2,000	$8,000
Mooring Frame	Steel/aluminum	4	$2,000	$8,000
Lab Analysis (year 1)	—	—	$25,000	$25,000
Data Integration	API setup, harmonization	—	$15,000	$15,000
Subtotal				**$196,600**
Contingency (30%)				**$58,980**
TOTAL				**$255,580**

8 M7: PAP DATA BACKBONE

8.1 7.1 Data Chain

Step	Action	Technology	Output
1	Collection	Physical collection	Raw sample
2	Spectroscopy	FTIR Spectrometer	Polymer signature
3	Fingerprinting	GC-MS or HPLC	Chemical fingerprint
4	Backtracking	Current models (SCUD, HYCOM)	Source vector
5	Attribution	Database lookup	Named entity
6	Publication	PAP Database	Open access data

8.2 7.2 Database Schema

Field	Type	Description
sample_id	UUID	Unique identifier
timestamp	DateTime	Collection time (UTC)
latitude	Float	GPS location
longitude	Float	GPS location
depth	Float	Collection depth
collection_method	Enum	Skimmer, Gandr, Bio-Bubble, SSS
polymer_type	Enum	PE, PP, PET, PS, PVC, Other
additive_fingerprint	JSON	Colorants, flame retardants
mass_mg	Float	Mass of sample
source_vector	Geography	Back-traced origin
matched_entity	String	Company name (if confirmed)
verification_status	Enum	Unconfirmed, Verified, Disputed

8.3 7.3 Public Dashboard Features

Feature	Description
Heat Map	Color-coded pollution density
Clickable Pins	Source attribution details
Timeline Slider	View pollution trends over time
Company Attribution List	Ranked by verified matches
Data Download	CSV, JSON, API endpoint
Data Request Portal	Custom reports for researchers

8.4 7.4 Data Release Strategy – Strategic Gradualism

Data release is treated as a **strategic asset**, not an activist weapon. The goal is to build scientific credibility, protect the project from legal attack, and maximize impact.

Phase	Data Status	Public Availability
Phase 0	Internal validation data	Internal team + advisors
Phase 1 (Year 3-4)	Peer-reviewed methodology paper	Published in scientific journal
Phase 1 (Year 4-5)	Aggregated regional data (no company attribution)	Public summary reports
Phase 1 (Year 5+)	Company-level data (with uncertainty bounds)	Public dashboard
Phase 2+	Full open data (peer-reviewed, contextualized)	Public dashboard + API

Key Principles:

- **No raw data is published without context.** All data is peer-reviewed, contextualized, and includes uncertainty bounds.
- **Companies are offered private pre-release review.** They can correct errors before publication.
- **Company-level data is released only after methodology is peer-reviewed.** This prevents attacks on credibility.
- **The "Shock and Awe" drop is replaced with "Strategic Gradualism."** Gradual, contextualized release builds credibility; sudden drops invite attacks.

8.5 7.5 PAP Negotiation Protocol

PAP is a **negotiation tool**, not a weapon. The goal is to convert polluters into funders and partners.

Phase	Action	Strategic Purpose
Year 3-4	Quiet outreach to top 20 identified polluters	"We have data. We would prefer to work with you."
Year 4-5	Private partnership agreements (if engaged)	"We will delay publication if you fund our operations and work on packaging redesign."
Year 5+	Public release (if refused)	"We offered them a chance to do the right thing. They refused."
Phase 2+	Continuous engagement	Ongoing partnerships with companies who engage; publication for those who don't

Key Principles:

- **Private outreach comes first.** Companies are given the opportunity to engage before public release.
- **Partnership agreements are the goal.** Funded partnerships are more valuable than public shaming.
- **Public release is the fallback, not the first move.** If companies refuse to engage, the data is published with full scientific rigor.
- **The moral high ground is maintained throughout.** The project offered cooperation; the company chose otherwise.

8.6 7.6 Threat Response Protocol

Escalation Level	Trigger	Action
Green	Normal operations	Continue silent data collection
Yellow	Legal threat or company hostility	Accelerate third-party verification; seek legal counsel
Orange	Coordinated lobbying or regulatory pressure	Activate legal defense fund; engage media partners
Red	Direct threat to personnel or infrastructure	Activate whistleblower protocols; publish data immediately

8.7 7.7 Data Hosting (Non-US)

Jurisdiction	Provider	Purpose	Rationale
Iceland	atNorth	Primary storage	EU jurisdiction; not subject to US CLOUD Act
Switzerland	Artmotion / Safe Swiss Cloud	Mirror backup	Strongest legal firewall against foreign interference

8.8 7.8 Phase 0 Cost

Component	Specification	Quantity	Unit Cost	Total
Database Infrastructure	Cloud (AWS/GCP)	1	\$5,000	\$5,000
API Development	Ingestion + Public API	400 hrs	\$100/hr	\$40,000
Dashboard Development	Heat map + UI	300 hrs	\$100/hr	\$30,000
FTIR Spectrometer	Portable, for Skimmers	1	\$40,000	\$40,000
GC-MS or HPLC	Lab analysis	1	\$50,000	\$50,000
External Data Integration	APIs, harmonization	80 hrs	\$100/hr	\$8,000
Subtotal				**\$173,000**
Contingency (30%)				**\$51,900**
TOTAL				**\$224,900**

9 M8: GOVERNANCE & TRUST

9.1 8.1 Trust Structure

Board of Trustees (7-9 members):

- 5-year staggered terms (max 2 terms)
- Extendable by Citizens' Assembly majority vote
- Emergency removal by petition + majority vote
- Authority: Policy, budget, Director appointment

Trust Director (1, appointed):

- 3-year term (renewable once)
- Authority: Day-to-day operations, staff hiring, budget execution

Science Advisory Council (5-7 members):

- 3-year terms (renewable once)
- Peer-elected from oceanographic/engineering institutions
- Authority: Technical advice, methodology review, data validation

Citizens' Assembly (50-100 members):

- 2-year terms (rotating 50%)
- Random stratified lottery (global + coastal communities)
- Authority: Binding recommendations on major decisions (75% super-majority for constitutional changes)

Operational Teams:

- Engineering, Data, Field Operations, Legal, Finance
- Hired by Trust Director; execute day-to-day work

9.2 8.2 Founder's Abdication

Phase	Role	Authority	Compensation
Phase 0	Founding Visionary & Principal Architect	Active	$90k-120k/year
Phase 1+	Formal Advisor	Advisory only (quarterly)	Class B share dividends
All	Emeritus Founder	Zero executive authority	3% Class B shares

9.3 8.3 Constitutional Provisions

Article	Provision	Ratification Requirement
I	Purpose and Mission	Unanimous
II	Trust Structure	75%
III	Founder's Abdication	75%
IV	Conflict of Interest Protocol	75%
V	Decision-Making Processes	75%
VI	Dispute Resolution	75%
VII	Amendment Process	75%
VIII	Financial Transparency & Auditing	75%
IX	Rights of Citizens	75%
X	Emergency Powers	75%

9.4 8.4 Conflict of Interest Protocol (Founder-Specific)

Clause	Implementation
Zero executive authority	Per Founder's Abdication
No special access	Proposals through standard process
Blind review	Independent panel; no knowledge of proposer
Board approval	Super-majority (75%)
Public disclosure	Amount, purpose, review process published
Recusal	Not participate in discussions or votes

9.5 8.5 Legal Shield Strategy

Project Jörmungandr operates in a high-risk legal environment. The PAP Protocol creates adversaries with deep pockets. The following legal protections are mandatory.

8.5.1: Trust Incorporation

Decision	Rationale
The Oceanic Public Trust shall be incorporated in Switzerland.	Switzerland offers the strongest legal protection for public interest research. Strict libel laws protect the project from defamation claims. Swiss courts do not recognize US libel judgments.

8.5.2: Vessel Flagging

Decision	Rationale
All Project Jörmungandr vessels shall be flagged in Panama or Liberia.	These jurisdictions do not recognize US libel judgments. Vessels in international waters are subject to flag state laws, not US laws.

8.5.3: Legal Defense Fund

Decision	Rationale
A $5-10M legal defense fund shall be established before Phase 1 operations begin.	The project will be sued. A legal defense fund ensures the project can survive prolonged legal challenges.

8.5.4: Legal Team

Decision	Rationale
A full-time international legal team shall be hired before Phase 1 operations begin.	Expertise is required in admiralty law, defamation defense, and public interest litigation. A retainer is insufficient.

9.6 8.6 Legal Authorization (UNCLOS)

Project Jörmungandr operates under the framework of **UNCLOS Article 87 (Freedom of Scientific Research)**. All operations in international waters are conducted as a scientific research expedition, which is explicitly protected under international law.

Additionally, the project seeks diplomatic endorsement from a host nation (e.g., the United States, Canada, New Zealand, or a Pacific Island nation) to provide diplomatic cover for operations in international waters. This is consistent with standard practice for ocean research vessels and scientific expeditions.

The legal team shall maintain documentation of the project's scientific research status, including methodology, data collection protocols, and publication plans, to demonstrate compliance with UNCLOS Article 87.

9.7 8.7 Phase 0 Governance (Provisional Entity)

Phase 0 is governed by a **Provisional Entity** established for the sole purpose of validating the technology. This entity may take the form of a non-profit corporation, a fiscal sponsorship, or a simple legal shell sufficient to receive funds and sign contracts.

The Provisional Entity is accountable to the founding consortium, consisting of Treg Robert Wells (Founding Visionary) and initial funders. All Phase 0 decisions are made by the founding consortium.

At the Phase 0→Phase 1 transition, the Provisional Entity dissolves, and the permanent Oceanic Public Trust structure (as defined in M8.1) is activated. All Phase 0 assets, data, and intellectual property are transferred to the Trust.

10 M9: PHASING & BUDGET

10.1 9.1 Consolidated Budget Summary

Note on Phase 2 Estimates: Phase 2 estimates are conceptual forecasts, not committed budgets. They represent a plausible scaling pathway based on Phase 0 and Phase 1 results. Actual Phase 2 scope, timeline, and cost will be determined during Phase 0-1 based on real-world data, operational experience, and funding availability. The project is not committing to a $6.6B Phase 2 before Phase 0 is validated.

Phase	Duration	CAPEX (Low)	CAPEX (High)	OPEX (Total)	Total (Low)	Total (High)
Phase 0	3 years	~$6.5M	~$8.8M	~$3.9M	**~$10.4M**	**~$12.7M**
Phase 1	5 years	~$58M	~$105M	~$31M	**~$89M**	**~$136M**
Phase 2 (Forecast)	10+ years	$6.6B	$13.2B	TBD	**$6.6B+**	**$13.2B+**

10.2 9.2 Revenue Model (Validated – Supplementary Revenue)

Revenue Stream	Phase 0	Phase 1 (Annual)	Phase 2 (Annual)	Notes
Recycled Plastics	$0	$3-8M	$15-40M	$500-1,000/ton (market reality)
Recycled Metals	$0	$0.5-1M	$5-10M	Pb, Cd, rare earths
Carbon Credits (Blue Carbon)	$0	$0.5-1.5M	$5-15M	~$5-15/ton (realistic)
Sustainability Stamp	$0	$2-10M	$20-50M	Primary economic engine
Donations/Grants	$3-5M	$10-20M	$25-50M	Foundations, UN, governments
TOTAL	**$3-5M**	**$16-40.5M**	**$70-165M**	

10.3 9.3 Operational Cost Offset Strategy

Revenue is treated as an **offset to operational costs**, not a profit center. The goal is to reduce the burden on public and philanthropic funding, not to achieve profitability.

Phase	Revenue Source	Expected Annual Offset	% of Annual OPEX Offset	Notes
Phase 0	Grants/Donations	\$3-5M	100%	Fully funded by philanthropy
Phase 1	Material sales, Carbon credits, Stamp	\$5-15M	15-30%	Early revenue generation
Phase 1 (Late)	Material sales, Carbon credits, Stamp	\$15-30M	40-60%	Scaling revenue
Phase 2	Material sales, Carbon credits, Stamp	\$30-75M	50-70%	Mature operations
Phase 2 (Mature)	Material sales, Carbon credits, Stamp	\$75-150M+	70-85%	**Significantly reduced public funding burden**

Key Principles:

- Revenue **reduces** the need for external funding but does not **replace** it
- The project remains a public utility with permanent public/philanthropic backing
- Material sales are **supplementary**, not foundational
- The Sustainability Stamp is the most promising revenue source for long-term cost offset

Phase 0–1 Funding Sources:

Source	Percentage	Rationale
Government/UN grants	50-60%	Stable, multi-year funding
Philanthropic foundations	20-30%	Rockefeller, Bezos Earth Fund
Corporate Sustainability Partnerships	10-15%	Direct contributions, not market-based
Material sales + Carbon credits	5-10%	Pure bonus, not budgeted

10.4 9.4 Phase Transition Triggers

Phase 0 → Phase 1	Phase 1 → Phase 2
≥2 strains >20% microplastic removal	>30% microplastic reduction
100+ successful Gandr missions	pH stabilization
>10kg heavy metals extracted	Oxygen stabilization
PAP dashboard functional	>500 tons plastics/year
≥\$50M secured for Phase 1	>100 companies identified
Constitution ratified	>40% operational cost offset

10.5 9.5 Timeline (Consolidated)

Activity	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9+
Phase 0 - Legal Setup	•								
Phase 0 - Engineering	•	•							
Phase 0 - Testing		•	•						
Phase 1 - Deployment Prep			•	•					
Phase 1 - Open-Ocean Ops				•	•	•			
Phase 1 - PAP Launch						•			
Phase 1 - Data Validation						•	•		
Phase 2 - Global Scaling									•

10.6 9.6 Key Milestones

Milestone	Year	Description
Trust Established	1	Legal incorporation; Constitution drafted
Phase 0 Deployment	1	All systems operational (coastal)
Phase 0 Data Complete	3	All strains validated; top performers selected
Phase 1 Funding Secured	3	≥$50M secured
Phase 1 Launch	4	Systems deployed in North Pacific Gyre
PAP "Strategic Gradualism" Release	6	Gradual, contextualized data release begins
Phase 1 Exit Criteria Met	8	Transition to Phase 2
Phase 2 Launch	9	Infrastructure build begins

11 M10: R&D CONTINUITY

11.1 10.1 R&D Department Structure

Team	Function	Phase 1 Staff	Phase 2 Staff
Bio-Bubble R&D	Strain refinement, mesh materials	2	5
Gandr R&D	Drone design, autonomy upgrades	2	5
SSS R&D	Deep-sea systems, umbilical reliability	0 (deferred)	4
PAP R&D	Data integration, dashboard evolution	1	3
Monitoring R&D	Sensor development, telemetry reliability	1	3
Operations Support	Field testing, deployment coordination	2	5
Administration	Budget, logistics, reporting	1	2
TOTAL		**9**	**27**

11.2 10.2 Staff Transition (Phase 0 → Phase 1)

Commitment	Implementation
All Phase 0 staff offered Phase 1 positions	Guaranteed offer letter at transition
No involuntary separation during transition	Performance-based reviews only
Salary continuity	No gap between Phase 0 and Phase 1
Benefits continuity	Health, retirement, Trust participation

11.3 10.3 Knowledge Transfer

Mechanism	Frequency	Purpose
Documentation	Continuous	Engineering specs, operations manuals
Training	Annual	New staff onboarding
Mentorship	6-12 months	Phase 0 → new hire knowledge transfer
Cross-Training	6-month rotations	R&D ↔ Operations

11.4 10.4 Continuous Improvement Cycle

Quarter	Focus	Activities
Q1	Review + Planning	Analyze data; set priorities
Q2	Design + Prototype	Develop improvements
Q3	Testing	Field test improvements
Q4	Implementation	Deploy to operational systems

11.5 10.5 Innovation Pipeline

Stage	Duration	Deliverable
Concept	6 months	Feasibility study
Prototype	6-12 months	Working prototype
Field Test	6-12 months	Field-validated system
Deployment	6-12 months	Integrated into operations

11.6 10.6 Priority Research Areas

Area	Phase 1 Focus	Phase 2 Focus
Strains	Optimize growth/uptake	Genetic enhancement
Bio-Bubble	Durability, mooring	Self-healing mesh
Gandr	Battery life, autonomy	Swarm logic, AI
SSS	(deferred)	Full autonomy
Monitoring	Sensor cost reduction	New metrics
PAP	Data integration	Real-time attribution

11.7 10.7 Founder's Role (Treg)

Aspect	Detail
Title	Founding Visionary, Emeritus
Phase 0	Active (full-time)
Phase 1+	Formal Advisor (quarterly check-ins, annual strategy review)
Compensation	Phase 0: $90k-120k/year; Phase 1+: Class B share dividends
Authority	Zero executive power (per Founder's Abdication)
IP Legacy	Original concepts = Treg; Improvements = Trust
Attribution	"Based on original concept by Treg Robert Wells"

11.8 10.8 R&D Budget

Phase	Annual Budget (Low)	Annual Budget (High)
Phase 1	$2.0M	$3.5M
Phase 2	$5.5M	$9.8M

12 M11: SUSTAINABILITY STAMP

12.1 11.1 Purpose

The Sustainability Stamp is the project's primary economic engine. It is a certification product sold to companies for the right to use "Jörmungandr Certified Ocean-Positive" branding on their products. Revenue from the Stamp reduces the burden on public funding and provides a sustainable cost offset.

12.2 11.2 Certification Standard

Element	Specification
Material Source	Plastic must be verifiably recovered from the ocean
Chain of Custody	Full audit trail from collection to recycling
Recycling Process	Verified environmental standards
Annual Reporting	Licensees must report on total ocean plastic usage

12.3 11.3 Audit Process

Step	Action	Frequency
1	Licensee submits materials for verification	Annual
2	Independent auditor reviews chain of custody	Annual
3	Project Jörmungandr certifies compliance	Annual
4	Licensee receives certification + branding rights	Annual

12.4 11.4 Fee Structure

Tier	Annual Fee	Benefits
Silver	$50,000	Use of "Certified" logo, basic reporting
Gold	$250,000	Use of "Certified" logo, PR support, annual impact report
Platinum	$1,000,000	Use of "Certified" logo, PR support, quarterly impact reports, Board observer rights

12.5 11.5 Target Licensees

Target	Category	Rationale	Status	Status Update
Unilever	Consumer goods	Desperate for genuine sustainability credentials	Target (no engagement yet)	—
P&G	Consumer goods	As above	Target (no engagement yet)	—
Nestlé	Food & Beverage	As above	Target (no engagement yet)	—
Coca-Cola	Beverage	As above	Target (no engagement yet)	—
PepsiCo	Beverage	As above	Target (no engagement yet)	—

Engagement Timeline:

Phase	Action	Responsible
Phase 0, Year 1	Initial outreach; introduction to project	MMEIA Director + Project Lead
Phase 0, Year 1-2	Present Sustainability Stamp concept; share MMEIA framework	MMEIA Director + Project Lead
Phase 0, Year 2-3	Negotiate early adopter agreements	Trust Board
Phase 1, Year 1	Formal launch of Sustainability Stamp	Trust Board
Phase 1, Year 2+	Expand to additional licensees	Trust Board

12.6 11.6 Revenue Projection

Phase	Annual Revenue	Notes
Phase 1 (Early)	\$0.5-2M	Early adopters
Phase 1 (Late)	\$5-10M	Scaling
Phase 2	\$20-50M	Mature product
Phase 2+	\$50-100M	Global adoption

13 M12: MARINE MAMMAL & ECOLOGICAL IMPACT ASSESSMENT (MMEIA)

13.1 12.1 Purpose & Rationale

Project Jörmungandr operates in the marine environment—the habitat of whales, dolphins, seals, sea turtles, and countless other species. While the project's goal is to remove pollution, the systems themselves must not become a new source of harm.

Core Principle: The project must demonstrate that its ecological benefits outweigh its operational risks. This requires rigorous assessment, monitoring, and mitigation.

Entanglement in fishing gear and marine debris is a global threat affecting at least **76% of pinniped species** and is a primary cause of human-caused mortality in many whale species, including right whales, humpback whales, and gray whales. Project Jörmungandr must not add to this problem.

13.2 12.2 Regulatory Framework & Compliance

Regulation	Applicability	Compliance Requirement
Marine Mammal Protection Act (MMPA)	All US waters	Authorization for incidental take; reporting requirements
Endangered Species Act (ESA)	All US waters	Consultation for listed species
UNCLOS Article 87 & 192	International waters	Freedom of scientific research; duty to protect and preserve marine environment
IMO Guidelines	International waters	Vessel standards; noise; pollution
Convention on Biological Diversity	International waters	Ecosystem-based management approach

13.3 12.3 Entanglement Risk Assessment – By Component

Component	Entanglement Risk	Risk Mechanism	Mitigation Strategy
Bio-Bubbles	Low-Medium	Mesh entanglement; mooring/tether lines	Passive mesh; slow drift; marine mammal observers; low-tension mooring; acoustic pingers
Gandr Drones	Low	Moving object collision; tether entanglement	Slow speed (2 knots); depth range (10-200m) where most large whales don't frequent; acoustic homing
Skin Skimmers	Medium	Trawl net entanglement	Modified nets with escape routes; slow tow speed; acoustic pingers; real-time monitoring
SSS Crawlers	Low	Umbilical entanglement on seafloor	Benthic operation; minimal surface expression; weighted, low-profile umbilical
Support Vessels	Medium	Vessel strike; anchor/-mooring lines	Vessel strike avoidance protocols; protected species observers; speed restrictions

13.4 12.4 Mitigation Measures – By Component

12.4.1 Bio-Bubbles

Mitigation	Description	Rationale
Passive Design	No active mechanical parts; soft mesh; moves with currents	Reduces risk of traumatic entanglement
Low-Tension Mooring	Minimal vertical lines; buoyant tethers	Reduces line entanglement risk
Acoustic Pingers	High-frequency acoustic deterrents	Alerts cetaceans to the presence of the structure
Marine Mammal Observers (MMOs)	Visual monitoring during deployment and maintenance	Detects animals in vicinity; enables avoidance

12.4.2 Skin Skimmers

Mitigation	Description	Rationale
Slow Trawling Speed	2 knots (2.8 km/h)—pace of a slow walk	Allows marine mammals to detect and avoid the net
Built-in Escape Routes	Modified net design with escape openings	Precludes trapping large animals
Acoustic Pingers	High-frequency deterrents on all nets	Alerts cetaceans to net presence; proven to reduce bycatch >90% when properly maintained
Real-Time Monitoring	Underwater cameras in collection zone	Enables immediate detection of incidental catch
Environmental Observers	Onboard monitoring during all operations	Enables immediate response to marine life sightings

The Ocean Cleanup's operational data provides a benchmark: over **8,300 observations of large marine animals** (megafauna), with **almost 5,000 observations of marine mammals**, most of which were not impacted by the system. Their incidental bycatch ratio was **0.6%** —over 99% of catch was plastic.

12.4.3 Gandr Drones

Mitigation	Description	Rationale
Depth Range (10-200m)	Operates below surface, above deep-diving whale range	Minimizes overlap with large whales
Slow Speed	2 knots cruising speed	Allows avoidance
Acoustic Homing	Emits detectable signal	Enables marine life to avoid the drone
Collision Avoidance	Forward-facing sonar	Detects and avoids large objects

12.4.4 SSS Crawlers

Mitigation	Description	Rationale
Benthic Operation	Operates on seafloor	Minimal surface interaction
Low-Profile Umbilical	Weighted, neutral buoyancy	Minimizes vertical lines in water column
Sediment Plume Management	Discharge diffuser returns water at depth	Minimizes benthic disturbance

13.5 12.5 Ecological Impact Assessment – By System

System	Potential Ecological Impact	Assessment Method	Mitigation
Bio-Bubbles	Algal bloom effects; thermal impact; habitat alteration	Water quality monitoring; phytoplankton surveys; thermal sensors	Density control; harvest protocols; kill switch
Skin Skimmers	Neuston bycatch; microplastic removal effects	Incidental catch analysis; neuston surveys	Slow speed; escape routes; monitoring
Gandr Drones	Mid-water bycatch; noise pollution	Acoustic monitoring; bycatch analysis	Depth restriction; acoustic mitigation
SSS Crawlers	Benthic habitat disturbance; sediment plume	Benthic surveys; sediment analysis	Slow operation; hotspot targeting; plume management
Support Equipment	Noise; vessel strike; fuel pollution	Vessel monitoring; noise logging	Speed restrictions; MMOs; fuel standards

13.6 12.6 Monitoring & Reporting Protocol

12.6.1 Marine Mammal Observation Protocol

Requirement	Implementation
Visual Monitoring	MMOs on all active trawling vessels during operations
Passive Acoustic Monitoring (PAM)	Hydrophones to detect cetacean vocalizations
Pre-Operation Scan	30-minute observation before deployment
Move-On Rule	If marine mammal sighted within 500m, pause operations
Incidental Catch Documentation	All bycatch identified, counted, and weighed

12.6.2 Reporting Requirements

Report	Frequency	Recipient
MMO Logs	Daily	Project Data Team
Incidental Catch Report	Per extraction	Project Data Team + PAP Database
Quarterly Ecological Summary	Quarterly	Trust Board + Public
Annual MMEIA Report	Annual	Trust Board + Regulatory Bodies
Emergency Entanglement Report	Immediate (within 24 hours)	NOAA Fisheries

12.6.3 Entanglement Response Protocol

Step	Action	Responsible
1	Detection: MMO or sensor detects entanglement	Onboard Observer
2	Immediate Report: Contact NOAA Fisheries hotline: (866) 755-NOAA	Vessel Captain
3	Cease Operations: Pause all nearby operations	Vessel Captain
4	Assessment: Assess animal condition and entanglement severity	MMO / Trained Responder
5	Response: If trained and authorized, attempt disentanglement	NOAA Large Whale Disentanglement Certification (or equivalent) required
6	Documentation: Photograph, log, and report all details	MMO
7	Follow-Up: Monitor animal if released; track if tagged	Response Team

Entanglement Response Team Certification Requirements:

Certification	Issuing Authority	Validity	Renewal
Large Whale Disentanglement Certification	NOAA Fisheries / Authorized Training Provider	2 years	Recertification required
Small Cetacean & Pinniped Disentanglement	NOAA Fisheries / Authorized Training Provider	2 years	Recertification required
Marine Mammal First Response	NOAA Fisheries / Authorized Training Provider	3 years	Recertification required
Vessel-Based Emergency Response	US Coast Guard / NOAA	5 years	Recertification required

13.7 12.7 Personnel Requirements

12.7.1 Staffing Overview – Principle of Proportionality

MMO requirements are **activity-based, not vessel-based**. The level of monitoring must match the level of operational risk:

Activity	Risk Level	MMO Requirement	Rationale
Phase 0 Testing (coastal)	Low	Per-trip/risk-based	Intermittent; limited scale
Active Trawling (Skimmers)	Medium-High	2 MMOs per active vessel	Highest entanglement risk
Bio-Bubble Deployment	Low	Visual monitoring by trained crew	Stationary, passive structure
Gandr Drone Operations	Low	Acoustic monitoring only	Operates below whale strike depth
Support Vessels	Medium	Vessel strike avoidance protocols	MMOs not required on all support vessels

12.7.2 Staffing Structure

Role	Phase 0	Phase 1	Phase 2 (Standard)	Phase 2 (Optimized)	Qualifications
Lead Marine Biologist / MMEIA Director	1 (FT)	1 (FT)	1 (FT)	1 (FT)	PhD in marine biology; 10+ years exp.
Senior MMO	1 (FT)	2 (FT)	4 (FT)	4 (FT)	JNCC/ACCOBAMS/BOEM cert.; 5+ years exp.
Field MMOs (per active Skimmer)	0 (per-trip)	2 per Skimmer	2 per Skimmer	**1 per Skimmer + Camera**	NOAA-approved MMO/PSO certification
PAM Specialist	0.5 (FT)	1 (FT)	2 (FT)	2 (FT)	Acoustic data analysis expertise
Ecological Impact Analyst	0.5 (FT)	1 (FT)	2 (FT)	2 (FT)	Quantitative ecology; statistical modeling
Data Manager / Reporting	0.5 (FT)	1 (FT)	2 (FT)	2 (FT)	Database management; GIS
Entanglement Response Team	0 (contract)	3 (on-call)	6 (on-call)	6 (on-call)	NOAA Large Whale Disentanglement Cert.
MMEIA Support Staff	0.5 (FT)	1 (FT)	2 (FT)	2 (FT)	Administrative; logistics
Total FTEs	**4-5**	**8-9**	**15-16**	**11-12**	

Note: The optimized Phase 2 approach (1 MMO + camera system per vessel) could reduce annual MMO costs by approximately 50% while maintaining ecological monitoring coverage. The decision between standard and optimized staffing will be made during Phase 0-1 based on operational data and camera system performance.

12.7.3 MMO Coverage Requirements

Phase	Active Skimmers	MMOs Required (Standard)	MMOs Required (Optimized)	Coverage Days/Year	Total MMO Days (Standard)	Total MMO Days (Optimized)
Phase 0	0-2 (testing)	Per-trip	Per-trip	30	60	60
Phase 1	5	10 (2 per vessel)	10 (2 per vessel)	200	2,000	2,000
Phase 2	50	100 (2 per vessel)	**50 (1 per vessel + cameras)**	250	25,000	**12,500**

12.7.4 Staff Costs

Role	Annual Salary (Low)	Annual Salary (High)	Source
Lead Marine Biologist	$100,000	$140,000	Market rate
Senior MMO	$75,000	$95,000	
Field MMO (per diem)	$375/day	$404/day	
PAM Specialist	$80,000	$110,000	Market rate
Ecological Analyst	$70,000	$95,000	Market rate
Data Manager	$65,000	$85,000	Market rate
Entanglement Response (per operation)	$5,000	$10,000	
MMO Training/Certification	$150	$2,200	

13.8 12.8 Equipment & Infrastructure Costs

12.8.1 Monitoring Equipment

Equipment	Unit Cost	Quantity (Phase 1)	Total (Low)	Total (High)	Source
Acoustic Pingers (per net)	$500	40	$20,000	$40,000	Market rate
Passive Acoustic Monitoring (PAM) Buoy	$1,200	10	$12,000	$24,000	
Low-cost Hydrophone (DIY)	$50	20	$1,000	$2,000	
Commercial Hydrophone	$5,000	5	$25,000	$50,000	
Underwater Cameras (per vessel)	$3,000	10	$30,000	$60,000	Market rate
Data Loggers / Recorders	$2,000	5	$10,000	$20,000	
GPS / Tracking Tags (for response)	$500	20	$10,000	$20,000	Market rate
Disentanglement Equipment Kits	$5,000	3	$15,000	$30,000	
Training & Certification	$2,200	10	$22,000	$44,000	
Subtotal (Phase 1)			**$145,000**	**$290,000**	

12.8.2 Facility & Infrastructure

Item	Phase 0	Phase 1	Phase 2	Notes
MMEIA Office / Lab Space	$20,000	$50,000	$150,000	Annual rent/utilities
Data Storage / Cloud Infrastructure	$5,000	$15,000	$50,000	Annual
Vessel Modifications (MMO stations)	$25,000	$100,000	$500,000	One-time
Research Vessel Charter (surveys)	$50,000	$150,000	$500,000	Annual

13.9 12.9 Program Costs – By Phase

12.9.1 Phase 0 (Years 1-3)

Cost Category	Annual (Low)	Annual (High)	3-Year Total (Low)	3-Year Total (High)
Personnel (4-5 FTEs)	$450,000	$650,000	$1,350,000	$1,950,000
MMO Per Diem (60 days)	$22,500	$24,240	$67,500	$72,720
Equipment (one-time)	$75,000	$150,000	$75,000	$150,000
Facilities & Operations	$100,000	$150,000	$300,000	$450,000
Training & Certification	$22,000	$44,000	$22,000	$44,000
Baseline Ecological Surveys	$100,000	$200,000	$300,000	$600,000
Contingency (20%)	$154,000	$244,000	$462,000	$732,000
TOTAL	**$923,500**	**$1,462,240**	**$2,576,500**	**$3,998,720**

12.9.2 Phase 1 (Years 4-8)

Cost Category	Annual (Low)	Annual (High)	5-Year Total (Low)	5-Year Total (High)
Personnel (8-9 FTEs)	$900,000	$1,300,000	$4,500,000	$6,500,000
MMO Per Diem (2,000 days)	$750,000	$808,000	$3,750,000	$4,040,000
Equipment (one-time + replacement)	$145,000	$290,000	$145,000	$290,000
Facilities & Operations	$200,000	$300,000	$1,000,000	$1,500,000
Entanglement Response (10 ops/yr)	$50,000	$100,000	$250,000	$500,000
Ecological Monitoring & Analysis	$150,000	$250,000	$750,000	$1,250,000
Training & Certification (annual)	$10,000	$20,000	$50,000	$100,000
Contingency (20%)	$441,000	$614,000	$2,205,000	$3,070,000
TOTAL	**$2,646,000**	**$3,682,000**	**$12,650,000**	**$17,250,000**

12.9.3 Phase 2 (Years 9-18)

Cost Category	Annual (Low)	Annual (High)	10-Year Total (Low)	10-Year Total (High)
Personnel (15-16 FTEs)	$1,600,000	$2,400,000	$16,000,000	$24,000,000
MMO Per Diem (25,000 days)	$9,375,000	$10,100,000	$93,750,000	$101,000,000
Equipment (replacement)	$100,000	$200,000	$1,000,000	$2,000,000
Facilities & Operations	$500,000	$750,000	$5,000,000	$7,500,000
Entanglement Response (25 ops/yr)	$125,000	$250,000	$1,250,000	$2,500,000
Ecological Monitoring & Analysis	$300,000	$500,000	$3,000,000	$5,000,000
Contingency (20%)	$2,400,000	$2,840,000	$24,000,000	$28,400,000
TOTAL	**$14,400,000**	**$17,040,000**	**$144,000,000**	**$170,400,000**

13.10 12.10 Program Cost Summary – All Phases

Phase	Duration	Total Cost (Low)	Total Cost (High)
Phase 0	3 years	**$2.58M**	**$4.00M**
Phase 1	5 years	**$12.65M**	**$17.25M**
Phase 2	10 years	**$144.00M**	**$170.40M**
TOTAL PROGRAM	**18 years**	**$159.23M**	**$191.65M**

13.11 12.11 Phased Implementation Timeline

Phase	MMEIA Activity	Deliverable	Timeline
Phase 0	Baseline ecological surveys; risk assessment	MMEIA Baseline Report	Year 1
Phase 0	MMO hiring and training; protocol development	Training Certification	Year 1-2
Phase 0	Acoustic pinger testing; escape route validation	Equipment Certification	Year 1-2
Phase 0	Passive acoustic monitoring (PAM) deployment	Baseline acoustic dataset	Year 2-3
Phase 1	Full MMEIA implementation (monitoring, reporting)	Quarterly Reports	Year 4-8
Phase 1	Independent ecological review	Peer-Reviewed Publication	Year 6
Phase 2	Adaptive management based on Phase 1 data	Updated MMEIA	Year 9+
Phase 2	Global MMEIA framework expansion	Global MMEIA Protocol	Year 10+

13.12 12.12 Funding Sources

Source	Phase 0	Phase 1	Phase 2
Project Operating Budget	50%	40%	30%
NOAA Fisheries Grants	25%	20%	15%
Philanthropic Foundations	25%	20%	15%
Sustainability Stamp Revenue	0%	20%	40%

13.13 12.13 Public Transparency & Reporting

Action	Frequency	Format
Publish MMEIA Baseline Report	Phase 0, Year 1	Public document
Publish Quarterly Ecological Summaries	Quarterly	Public dashboard
Publish Incidental Catch Data	Per extraction	PAP Database
Publish Annual MMEIA Report	Annual	Public document
Publish Peer-Reviewed Ecological Review	Phase 1, Year 6	Scientific journal

13.14 12.14 Success Metrics

Metric	**Phase 0 Target**	**Phase 1 Target**	**Phase 2 Target**
MMO Observation Coverage	100% of high-risk operations	100% of active Skimmers	100% of active Skimmers
Incidental Bycatch Ratio	$<$1%	**$<$0.5%**	**$<$0.5%**
Entanglement Response Time	$<$4 hours	$<$2 hours	$<$1 hour
Ecological Baseline Data	Complete	Updated annually	Updated annually
Peer-Reviewed Publications	0	$\geq$1	$\geq$3
Regulatory Compliance	100%	100%	100%

Transparency Commitment: The incidental bycatch ratio will be tracked aggressively and published transparently as part of the quarterly ecological summaries (M12.6.2). Data will be made available through the PAP public dashboard (M7.3) to ensure accountability and continuous improvement. The target of $<$0.5% is ambitious but realistic, benchmarked against The Ocean Cleanup's operational ratio of 0.6%.

13.15 12.15 Risk Register – MMEIA

Risk	**Probability**	**Impact**	**Mitigation**	**Status**
Incidental marine mammal entanglement	Low	Critical	Acoustic pingers; escape routes; MMOs; slow speeds	**Addressed**
Regulatory non-compliance	Low	Critical	NOAA consultation; legal team; compliance officer	**Addressed**
Insufficient MMO staffing	Medium	High	Multiple training pipelines; competitive compensation	**To Plan**
Equipment failure (pingers, cameras)	Medium	Medium	Redundant systems; pre-deployment testing	**To Plan**
Baseline data gaps	Medium	Medium	Comprehensive Phase 0 surveys; expert consultation	**To Plan**
Entanglement response failure	Low	High	Redundant response teams; NOAA coordination	**To Plan**

COMPETITIVE LANDSCAPE

Project	Approach	Scale	Status	Differentiator
The Ocean Cleanup	Passive U-shaped booms	GPGP	Operational	Macroplastic removal
4ocean	Commercial collection + retail	Coastal	Operational	Consumer engagement
SeaBin	Harbor-based collection	Small-scale	Operational	Localized cleanup
Plastic Bank	Land-based recycling in developing countries	Scaled	Operational	Social enterprise
Project Jörmungandr	Multi-tier: Skimmers, Gandr, Bio-Bubbles, SSS	Global	Planned	**PAP Protocol (data product)**

Jörmungandr's Unique Advantage: The PAP Protocol creates a data product with political and legal utility. No other project does chemical fingerprinting + backtracking + public attribution at scale. This generates value beyond physical cleanup.

PARTNERSHIPS (Tracking)

Category	Target Partners	Purpose	Timeline
Research	Scripps, Woods Hole, NOAA, GEOMAR	Scientific credibility	Phase 0-1
Legal	Greenpeace, ClientEarth, Earthjustice	Political and legal firepower	Phase 0-1
Funding	Rockefeller Foundation, Bezos Earth Fund, UNEP	Financial backing	Phase 0-1
Corporate	Unilever, P&G, Nestlé, Coca-Cola, PepsiCo	Sustainability Stamp licensees	Phase 1-2

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CROSS-REFERENCE MAP

Concept	Primary Module	Also Appears In
Bio-Bubble System	M2	M1, M9, M12
Gandr Drone & Micro-Hive	M3	M1, M9, M12
SSS Crawler & Barge	M4	M1, M9, M12
Skin Skimmer	M5	M1, M9, M12
Monitoring & Telemetry	M6	M1, M7, M9
PAP Data Backbone	M7	M1, M6, M9
Governance & Trust	M8	M9, M10
Phasing & Budget	M9	All
R&D Continuity	M10	M8, M9
Sustainability Stamp	M11	M9
Marine Mammal & Ecological Impact Assessment	M12	M5, M6, M9

DOCUMENT CONCLUSION

Project Jörmungandr is a complete, validated, and modular oceanic regeneration architecture. It is designed to be:

- **Self-sustaining** – Closed-loop materials, energy, and logistics
- **Autonomous** – Minimal human intervention beyond high-level governance
- **Trigger-based** – Phased deployment triggered by measurable ecological metrics
- **Resilient** – Graceful failure, redundancy, low-tech overrides
- **Transparent** – Public Trust governance, open data, and accountability
- **Legally Protected** – Swiss incorporation, vessel flagging, legal defense fund, UNCLOS authorization
- **Economically Sustainable** – Public utility framing, Sustainability Stamp as primary offset
- **Ecologically Responsible** – Comprehensive MMEIA framework covering all components

The Big Picture:

- **Phase 0:** Validates core technology
- **Phase 1:** Demonstrates measurable ecological impact at basin-scale
- **Phase 2:** Scales systems to global deployment
- **Phase 3+:** The long-term vision for complete ocean regeneration

Next Steps: Phase 0 implementation begins with legal incorporation, site selection, engineering validation, and baseline ecological surveys.

"Plant a seed, nurture the tree, share the fruit of your labors."

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